

Hw

$$\begin{aligned} \text{Max } & x_p(2.253 - 0.013x_p + 0.003x_o) + x_o(1.142 - 0.015x_o + 0.005x_p) - 75y \\ & 2.253 - 0.013x_p + 0.003x_o \leq y \\ & 1.142 - 0.015x_o + 0.005x_p \leq y \\ & x_p, x_o, y \geq 0 \end{aligned}$$

Problem 1

Chapter 14/Problem 70 (p.707). In the FPL electricity pricing model, we assumed that the capacity level is a decision variable.

Assume now that capacity has already been set at 0.65 kWh. (Note that the cost of capacity is now a sunk cost, so it is irrelevant to the decision problem.) Change the model appropriately.

1(a) Write a mathematical programming model.

1(b) Solve this model

Problem 2 [Optional for LSCM]

2(a) Write the problem using the matrix notation

$$\max \{f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} : \mathbf{A} \mathbf{x} \leq \mathbf{b}\}$$

2(b) Is this problem convex quadratic program (CQP)?

2(c)[Optional] Solve this problem by hand.